Pitch canker in Europe

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1 Overview

Pitch canker, caused by the fungus *Fusarium circinatum*, attacks pines and Douglas fir, killing branches and sometimes whole trees. By the 2000s it had reached plantations and nurseries in northern Spain, Portugal, France and Italy, carried in seed and seedlings. Möykkynen et al. (2015) modelled how it could spread through Europe from its points of entry, on cells of 1 km in yearly steps. Infected seedling lots are carried from nurseries or ports to forests up to 300 km away and release spores where they are planted. An infected tree then spreads the fungus for five years before it dies, by spores carried a few hundred metres to its own and nearby cells, and by insects, mostly bark beetles, flying about a kilometre to another tree. Infection succeeds in proportion to the healthy trees in a cell and to the cell's climatic suitability, an ecoclimatic index from 0 to 1. The authors predicted spread to the pine forests of northern Spain and south-west

France, and argued that because spores fly so short a distance, the fungus "cannot easily cross spatial discontinuities in the distribution of host species".

This implementation is an adaptation. The source's European maps of host cover and climatic suitability were not published, and its program is available only from the authors, so the model runs on a synthetic coast. It reproduces the source's mechanisms and the directions of its sensitivity analyses, not its maps.

```
ENV["GKSwstype"] = "100"
using EcologicalCellularAutomata
using CellularAutomataCore
using Plots
using Statistics
```

2 The model

Each cell of 1 km² holds up to `N_max` = 500 trees per hectare × 100 ha = 50,000 host trees, in proportion to its host cover. Every infection, whatever carries it, takes one healthy tree with probability (Eq. 1 of the source)

`p = (N_healthy / N_max) × EI`.

Spores from a cell's spreading trees give a density `N_spreading / N_max` at the cell's midpoint, falling as `exp(-d / 280 m)` out to 1,300 m (Eqs. 2-3), and a cell gains `N_healthy × EI × density` infections from them (Eq. 4). On 1 km cells that reach covers only the cell itself and its four orthogonal neighbours, whose weight is small:

```
(neighbour_weight=round(exp(-1000 / 280); digits=4),
 diagonal_distance_m=round(Int, hypot(1000, 1000)))
```

```
(neighbour_weight = 0.0281, diagonal_distance_m = 1414)
```

The model is a cellular automaton for the spores and the ageing of infected trees, and a callback for the seedling lots and the insects, which act at a distance one tree at a time:

```
model = build_pitch_canker()
(layers=length(model.states), parameters=length(model.parameters))
```

```
(layers = 12, parameters = 17)
```

3 The synthetic coast

The coast runs along the left edge of a landscape 600 km long and 200 km deep. Host cover is patchy, half the cells holding none; the ecoclimatic index is 1 on a coastal strip 40 km wide and falls to 0 at 120 km, as the source's index is highest on the coasts. Three entry points on the coast stand in for Gijón and Bilbao, about 200 km apart, from the first year, and Brindisi from the fourth, each sending 100 infected seedling lots a year (Table 2 of the source).

```
prob = problem_pitch_canker(; seed=1)
u0 = prob.u0
entries = prob.p.entries
function panel(A, title; kw...)
    heatmap(A; aspect_ratio=:equal, xlabel="inland (km)", ylabel="along the coast
    (km)",
            title, titlefontsize=10, kw...)
end
plot(panel(u0.coverage, "host cover"; c=:Greens),
     panel(u0.ei_now, "ecoclimatic index"; c=:Oranges),
     layout=(1, 2), size=(700, 650))
scatter!([e.j for e in entries], [e.i for e in entries]; subplot=1, label="entry",
         c=:red, ms=5)
```

4 Twenty years from three entry points

```
sol = solve(prob, Synchronous())
area = [pitch_canker_infected_area(u) for u in sol.u]
u = sol.u[end]
(infected_area_percent=round.(area[[6, 11, 16, 21]]; digits=1),
 cells_infected=count(>(0), u.infected_year),
 trees_alive_infected=sum(pitch_canker_infected(u)), trees_dead=sum(u.dead))
```

```
(infected_area_percent = [21.8, 47.1, 54.5, 56.5], cells_infected = 33814,
trees_alive_infected = 324273460, trees_dead = 356202196)
```

```
year = Float64.(u.infected_year)
year[year .== 0] .= NaN
alive = pitch_canker_infected(u) ./ 100
alive = map(x -> x > 0 ? x : NaN, alive)
plot(panel(year, "year of infection"; c=:viridis),
     panel(alive, "infected trees per ha"; c=:Reds),
     layout=(1, 2), size=(700, 650))
```

Over 10 runs, the probability that each cell is infected within 20 years (the source's Fig. 6):

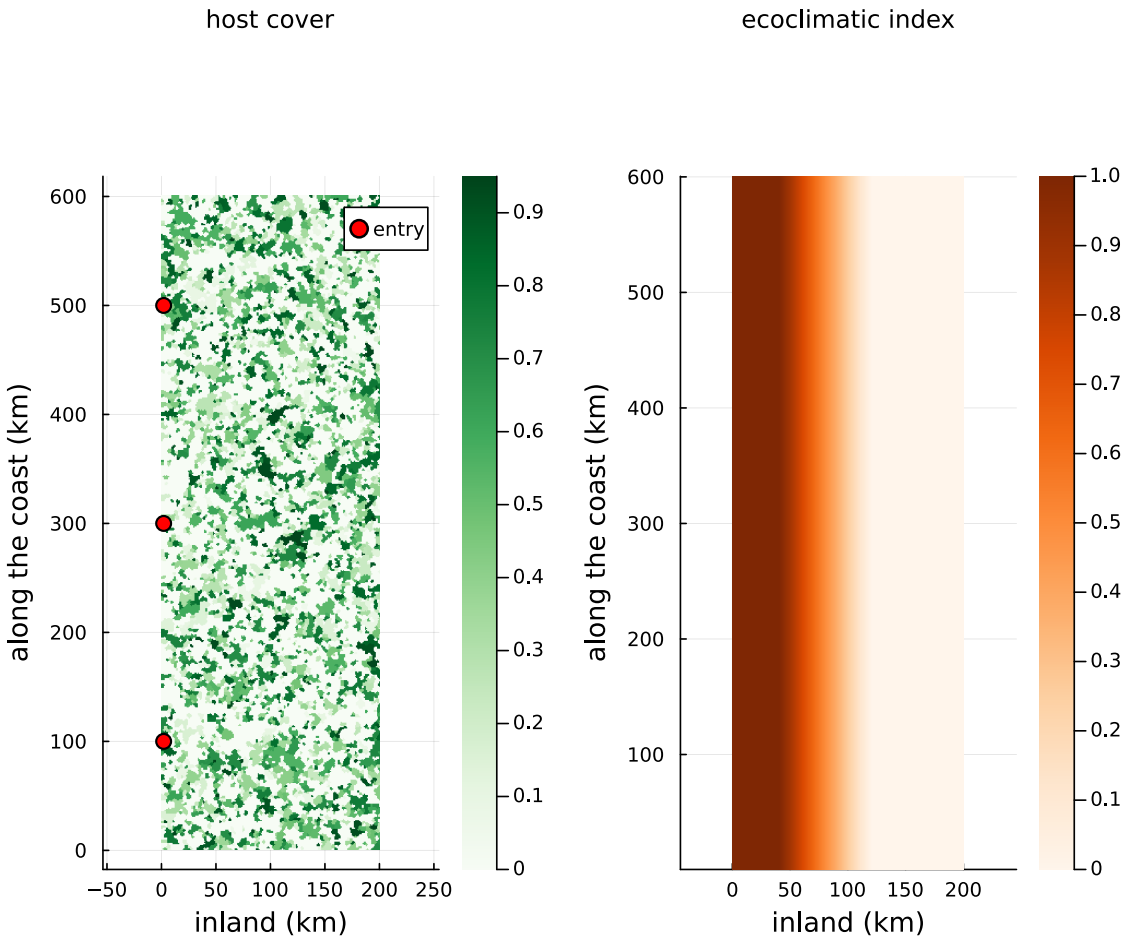


Figure 1: The synthetic coast: host cover (left) and ecoclimatic index (right), with the three entry points.

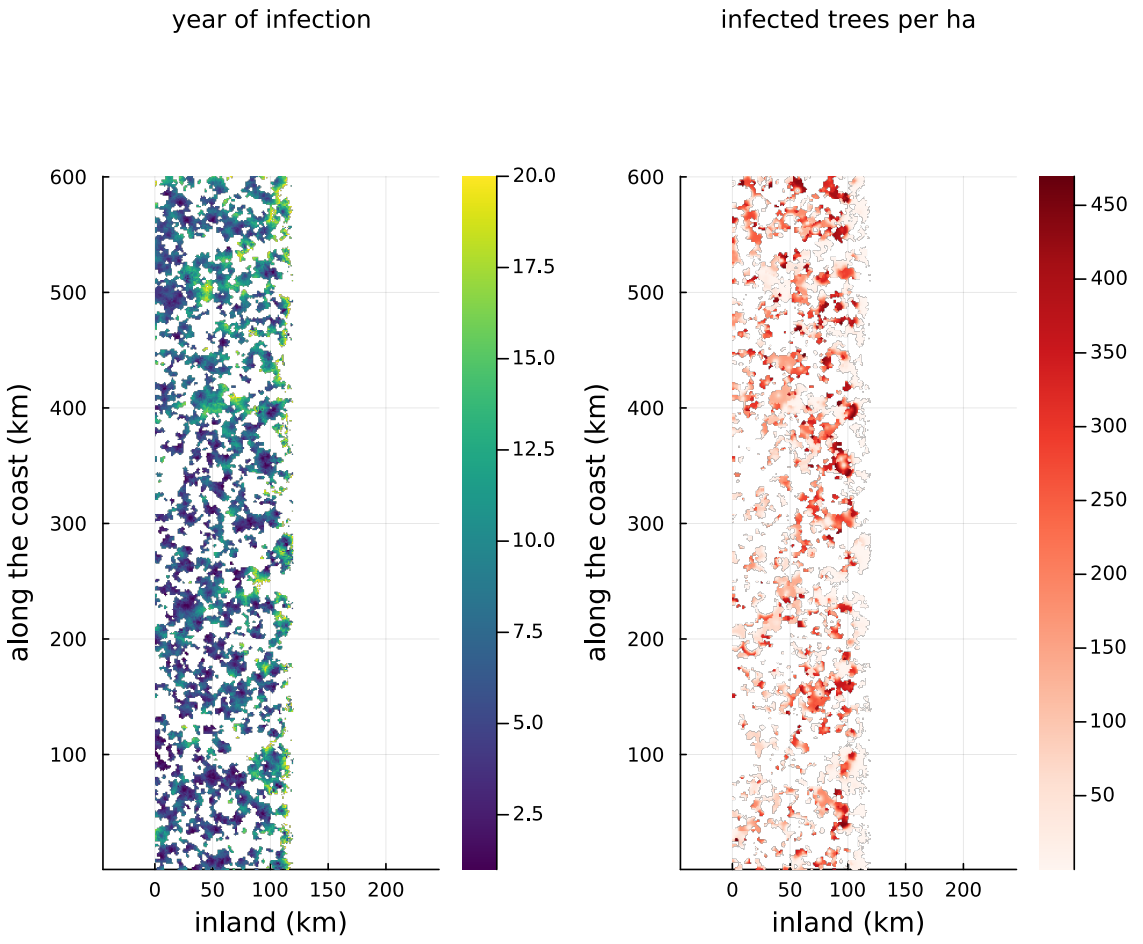


Figure 2: One run: the year each cell was first infected (left) and the infected trees per hectare still alive after 20 years (right).

```
runs = [solve(remake(prob; seed=s), Synchronous()).u[end] for s in 1:10]
P = sum(r -> r.infected_year .> 0, runs) ./ 10
P[u0.coverage .== 0] .= NaN
panel(P, "probability of infection"; c=:YlOrRd, size=(400, 650))
```

5 Sensitivity

The source changed one parameter at a time and followed the “percentage of infected host area” over 20 years, 10 runs each (its Fig. 8). The source does not define the measure; here it is the host area of the cells holding any infected or dead tree, as a share of all host area. Each run below draws its own landscape from its seed, the same ten for every setting.

```
# the share of host trees infected or dead, the other reading of the source's
measure
function tree_share(v)
    hit = sum(pitch_canker_infected(v)) + sum(v.dead)
    return 100 * hit / (sum(v.hosts) + hit)
end
function curve(; lots=100, spreading_years=5, pmap=(;), seeds=1:10)
    runs = map(seeds) do seed
        es = [merge(e, (; lots)) for e in pitch_canker_entries(600)]
        solve(problem_pitch_canker(; seed, entries=es, spreading_years, pmap),
              Synchronous()).u
    end
    m(f) = vec(mean(reduce(hcat, [f.(r) for r in runs]); dims=2))
    return (; area=m(pitch_canker_infected_area), trees=m(tree_share))
end
settings = [("seedling lots a year", [(; lots=l) for l in (10, 100, 1000)], ["10",
"100", "1000"]),
            ("spore dispatches per lot",
             [(; pmap=(; dispatches=d)) for d in (10, 100, 1000)], ["10", "100",
"1000"]),
            ("insects per infected tree",
             [(; pmap=(; insects_per_tree=k)) for k in (0.5, 1.0, 2.0)], ["0.5",
"1", "2"]),
            ("years a tree spreads", [(; spreading_years=y) for y in (1, 5, 10)],
             ["1", "5", "10"]),
            ("spore flight (mean/max, m)",
             [(; pmap=(; spore_mean=a, spore_max=b))
              for (a, b) in ((140.0, 650.0), (280.0, 1300.0), (560.0, 2600.0))],
             ["140/650", "280/1300", "560/2600"]),
            ("insect flight (mean/max, m)",
             [(; pmap=(; insect_mean=a, insect_max=b))
              for (a, b) in ((500.0, 2250.0), (1000.0, 4500.0), (2000.0, 9000.0))],
             ["500/2250", "1000/4500", "2000/9000"])]
results = [[curve(; s...) for s in set] for (_, set, _) in settings]
curves = [[r.area for r in rs] for rs in results]
```

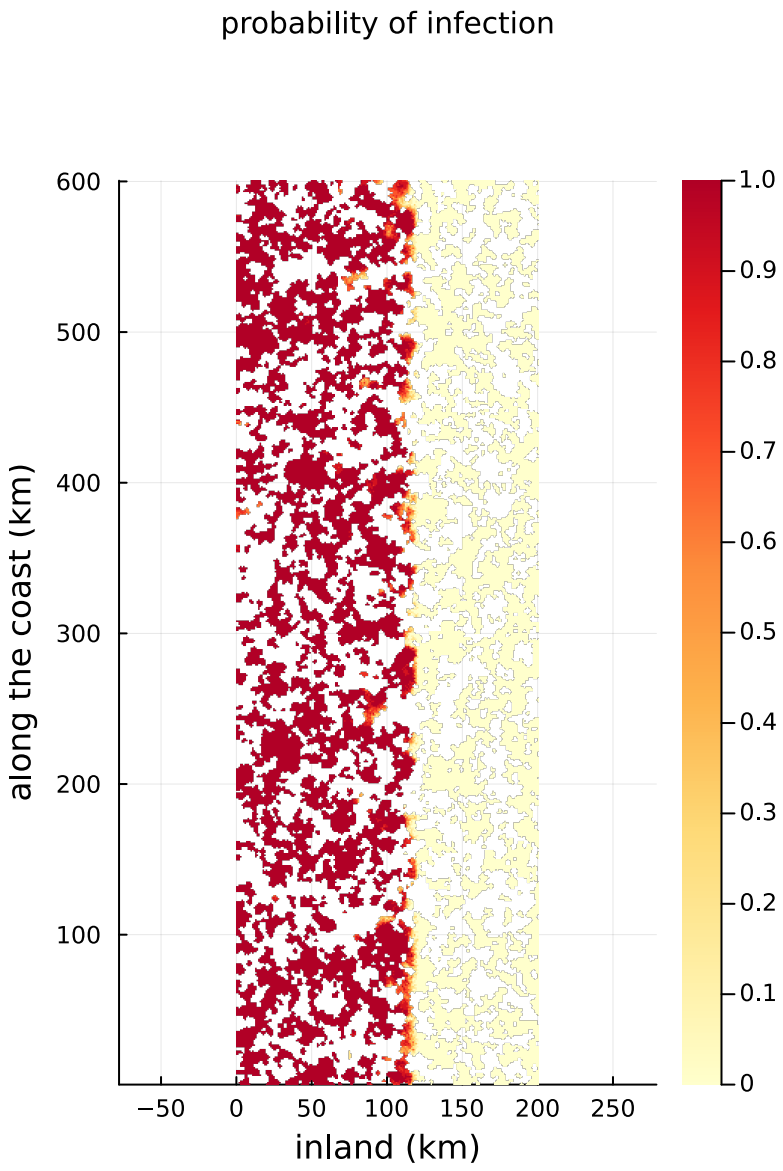


Figure 3: The probability that each cell is infected within 20 years, over 10 runs on the same landscape.

```
[(setting=name, at_year_10=round.(getindex.(c, 11); digits=1),
  at_year_20=round.(getindex.(c, 21); digits=1))
 for ((name, _, _), c) in zip(settings, curves)]
```

```
6-element Vector{@NamedTuple{setting::String, at_year_10::Vector{Float64},
at_year_20::Vector{Float64}}}:
 (setting = "seedling lots a year", at_year_10 = [15.4, 48.2, 57.9], at_year_20 =
 [43.0, 57.4, 59.2])
 (setting = "spore dispatches per lot", at_year_10 = [38.8, 48.2, 53.0], at_year_20
 = [55.0, 57.4, 58.6])
 (setting = "insects per infected tree", at_year_10 = [42.5, 48.2, 52.8], at_year_20
 = [55.4, 57.4, 58.8])
 (setting = "years a tree spreads", at_year_10 = [42.3, 48.2, 48.2], at_year_20 =
 [52.4, 57.4, 57.6])
 (setting = "spore flight (mean/max, m)", at_year_10 = [47.9, 48.2, 50.1],
 at_year_20 = [57.3, 57.4, 57.7])
 (setting = "insect flight (mean/max, m)", at_year_10 = [39.1, 48.2, 53.6],
 at_year_20 = [55.2, 57.4, 58.4])
```

```
plots = map(zip(settings, curves)) do ((name, _, labels), cs)
    p = plot(; title=name, titlefontsize=9, xlabel="year", ylabel="infected host
area (%)",
             legend=:bottomright, legendfontsize=7, ylim=(0, 65))
    for (c, l, s) in zip(cs, labels, (:dash, :solid, :dot))
        plot!(p, 0:20, c; label=l, ls=s, lw=2, c=:black)
    end
    p
end
plot(plots...; layout=(3, 2), size=(700, 850))
```

Four of the six parameters move the infected area the way the source found. Ten seedling lots a year spread the fungus far more slowly than 100, and 1,000 faster still (15, 48 and 58% at year 10), where the source found 100 and 1,000 nearly the same. More spore dispatches per lot spread it faster, and so, by a little, do more insects per tree. A tree that spreads for one year instead of five slows it, and ten years add nothing, as in the source. The two flight distances do not match. A longer spore flight changes almost nothing here (48 to 50%), where the source found it mattered, and a longer insect flight changes a lot (39 to 54%), where the source found no effect.

Both differences come from the measure. A cell counts as infected once it holds one infected tree, so anything that seeds new cells raises the area. Insects flying farther reach more cells, while spores, on cells of 1 km, never leave a cell and its four neighbours; a longer spore flight only weights those neighbours more. Measured instead as the share of host trees infected or dead, the same runs give:

```
[(setting=name, trees_at_year_10=round.([r.trees[11] for r in rs]; digits=1))
 for ((name, _, _), rs) in zip(settings, results)]
```

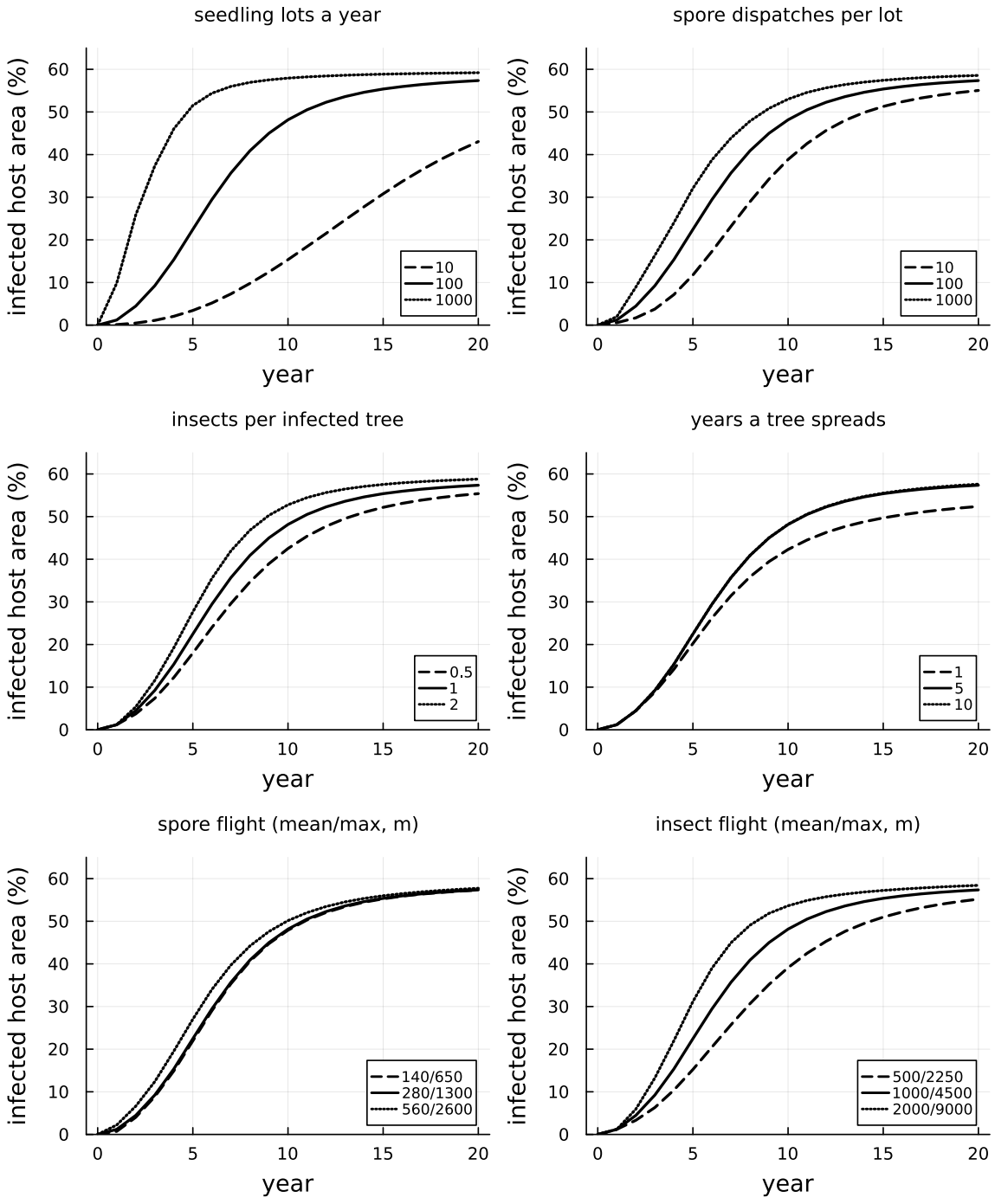


Figure 4: One-at-a-time sensitivity of the infected host area (mean of 10 runs), as the source's Fig. 8: the baseline value of each parameter is the middle line.

```
6-element Vector{@NamedTuple{setting::String, trees_at_year_10::Vector{Float64}}}:
 (setting = "seedling lots a year", trees_at_year_10 = [0.5, 4.4, 22.2])
 (setting = "spore dispatches per lot", trees_at_year_10 = [0.6, 4.4, 15.4])
 (setting = "insects per infected tree", trees_at_year_10 = [1.1, 4.4, 17.1])
 (setting = "years a tree spreads", trees_at_year_10 = [0.4, 4.4, 4.4])
 (setting = "spore flight (mean/max, m)", trees_at_year_10 = [3.9, 4.4, 11.5])
 (setting = "insect flight (mean/max, m)", trees_at_year_10 = [3.7, 4.4, 4.8])
```

With that reading the insect flight has almost no effect, as the source found, and the longest spore flight a large one, but the number of insects per tree then matters a great deal, which the source did not find, and 1,000 lots a year spread much faster than 100. Neither reading reproduces all six panels, and without the source's program which one it used cannot be settled.

The curves here also level off, near 58%, once the suitable strip of coast is full: almost every forest cell in the strip is infected in every run, and none beyond it (Figure 3). The source's curves, over the whole of Europe, are still rising after 20 years.

6 Gaps in the forest

The source's conclusion that the fungus cannot easily cross gaps in its hosts can be checked directly. Two blocks of forest, each 20 km wide, are separated by a gap without hosts; ten seedling lots are planted once, 3 km from the edge of the first block, and the question is whether, within 20 years, any tree of the second block is infected, with spores alone or with insects too:

```
function crossed(gap, insects; seed)
    m, n = 40, 40 + gap
    cov = [j <= 20 || j > 20 + gap ? 0.5 : 0.0 for _ in 1:m, j in 1:n]
    p = problem_pitch_canker(; coverage=cov, ei=ones(m, n), years=20, seed,
                             entries=[(; i=20, j=18, first=1, last=1, lots=10)],
                             pmap=(; transport_max=0.0,
                                   insects_per_tree=insects ? 1.0 : 0.0))
    v = solve(p, Synchronous()).u[end]
    return any(>(0), v.infected_year[:, (21 + gap):end])
end
[(gap_km=g, spores_only=count(s -> crossed(g, false; seed=s), 1:10),
  with_insects=count(s -> crossed(g, true; seed=s), 1:10)) for g in 0:5]
```

```
6-element Vector{@NamedTuple{gap_km::Int64, spores_only::Int64,
with_insects::Int64}}:
 (gap_km = 0, spores_only = 10, with_insects = 10)
 (gap_km = 1, spores_only = 0, with_insects = 10)
 (gap_km = 2, spores_only = 0, with_insects = 10)
 (gap_km = 3, spores_only = 0, with_insects = 10)
 (gap_km = 4, spores_only = 0, with_insects = 0)
 (gap_km = 5, spores_only = 0, with_insects = 0)
```

Spores alone never cross a gap of even 1 km: from a cell's midpoint they reach at most 1,300 m, which covers only the four neighbouring cells. Insects, flying up to 4,500 m, cross gaps of

up to 3 km in every run and never one of 4 km, whose far side lies 4.5 km from the midpoint of the last forest cell. In this model a strip of 4 km without hosts stops the fungus unless infected seedlings are carried across it, which is the source's conclusion: its spread can be stopped by isolating infected forests and preventing new arrivals.

7 Insect flight distributions

The source also compared an exponential distribution of insect flights with Weibull distributions of shape 1 and 1.5, all of mean 1,500 m, each with and without a cut at 4,500 m (its Fig. 9), and found little difference, the cut slowing the spread slightly. A Weibull distribution of shape 1 is the exponential, so two distributions remain:

```
fig9 = [("exponential", (; insect_distribution=:exponential)),
        ("Weibull, shape 1.5", (; insect_distribution=:weibull, insect_shape=1.5))]
[(distribution=name, cut=cut,
  at_year_10=round(curve(; pmap=merge(q, (; insect_mean=1500.0,
                                          insect_max=cut ? 4500.0 : Inf))).area[11];
                   digits=1))
 for (name, q) in fig9 for cut in (false, true)]
```

```
4-element Vector{@NamedTuple{distribution::String, cut::Bool, at_year_10::Float64}}:
 (distribution = "exponential", cut = 0, at_year_10 = 52.8)
 (distribution = "exponential", cut = 1, at_year_10 = 50.3)
 (distribution = "Weibull, shape 1.5", cut = 0, at_year_10 = 51.3)
 (distribution = "Weibull, shape 1.5", cut = 1, at_year_10 = 50.7)
```

As in the source, the shape of the distribution matters little (52.8 against 51.3% at year 10 without a cut), and cutting flights at 4,500 m slows the spread slightly (to 50.3 and 50.7%).

8 Summary

Claim of the source	Source	Here
The fungus spreads from its entry points into nearby forests where the climate suits it	maps of Europe (Figs. 6, 7)	yes: almost every forest cell where the climate allows infection is reached in every run, and none where it does not
Sensitivity of the infected host area (Fig. 8)	six panels	four of six directions match; the spore and insect flight distances do not (a different three match when the measure counts trees)

Claim of the source	Source	Here
The shape and cut of the insect flight distribution matter little (Fig. 9)	small differences, the cut slower	yes
Spores cannot cross gaps in the hosts	argued from the short spore flights	yes: spores cross no gap, insects up to 3 km
Maps of Europe, and spread under a changing climate over 100 years (Figs. 5, 10, 11)	maps	not repeated: the host and climate maps are not published

9 Deviations from the source

```
foreach(d -> println("- ", d), model_info(:pitch_canker).deviations)
```

```
- The source's European host rasters (European Forest Institute species maps) and
CLIMEX ecoclimatic index maps were not published with the article, and its program
(PitchCanker) is available only from the authors, so the model runs on a synthetic
coast of 600 x 200 cells (600 km along the coast, 200 km inland): patchy host
coverage from a random cluster landscape, half the cells without hosts, and an index
that is 1 on a coastal strip 40 km wide and falls to 0 at 120 km, with three entry
points on the coast in place of Gijón, Bilbao and Brindisi.
- The insects of each cell are drawn together (insect_method = :pooled): given the
healthy trees when they set out, the numbers infecting each nearby cell, over all
their tries, are a multinomial draw with the per-insect probabilities. This agrees
in distribution with flying each insect (insect_method = :each, as the source does)
as long as one cell's insects take few of any cell's healthy trees, and makes runs
fast when millions of trees spread.
- The source does not say whether a seedling lot infects a tree where it is planted;
here it infects only through its spore dispatches, as its Fig. 2 draws them.
- Flight distances above their maximum (1,300 m for spores, 4,500 m for insects) are
drawn again; the source does not say how the distributions are truncated.
- An insect that fails to infect flies again from its tree; a flight landing outside
the lattice or on a cell without healthy hosts is a failed try. The source does not
say where later tries start.
- Within a year the spores act first, then the seedling lots, then the insects, all
from the trees spreading at the start of the year, and each infection lowers the
healthy trees at once; the source gives no order. Spore infections from several
donor cells are summed before rounding.
- A fractional number of insects (0.5 per tree in Fig. 8c) is rounded at random cell
by cell.
- The infected host area is the host area of cells with any infected or dead tree,
as a percentage of all host area; the source does not define it.
- The ecoclimatic index is interpolated from the current climate in the first year
(2005) to the future climate in 2100; the source interpolates between its 1961-1990
and 2071-2100 maps without saying which year each stands for.
```

References

Möykkynen, Timo, Paolo Capretti, and Timo Pukkala. 2015. "Modelling the Potential Spread of *Fusarium Circinatum*, the Causal Agent of Pitch Canker in Europe." *Annals of Forest Science* 72 (2): 169–81. https://doi.org/10.1007/s13595-014-0412-2.